

Climate preconditioning and the mechanics of hazardous alpine mass movements: a catalog and attribution protocol

Bradley P. Lipovsky

Working manuscript, 1 September 2026

Abstract

Large rock, ice, and mixed-material collapses can transform into far-traveled debris flows or locally extreme tsunamis. Warming changes glacier geometry, permafrost temperature, meltwater supply, and precipitation phase. However, event reports seldom separate these preconditioning changes from immediate weather, earthquakes, structure, and progressive damage. Here, I define a climate-blind event inventory and an attribution protocol that preserves this distinction. Two structured frames contain 40 and 38 rows that cross an operational volume, travel, or runup prefilter. An exact provenance crosswalk gives every row a visible disposition, and supplemental searches extend the screen to 148 candidate occurrences. Two independent screens include 73, retain 68 as uncertain, and exclude 7; 53 included failures have day-scale or better timing for a later trigger analysis. Trigger, cascade, and repeat-site dependence remain separate. A registered ERA5 pilot then retains the 29 of 53 included, trigger-time-eligible occurrences that have finite catalog coordinates. For the registered seven-day pre-date window, the median rank among 1991–2020 calendar controls is 0.586; after a robust linear trend is removed, the median residual rank is 0.483. A conservative-antecedence sensitivity gives 0.586 and 0.533. The median four-cell rank ranges are 0.133 and 0.103, respectively. An independently reviewed source-coordinate and UTC-onset audit subsequently accepts source coordinates for 24 occurrences, UTC intervals for 34, and both for 22; only 9 of the pilot’s 29 occurrences have accepted source coordinates. The registered pilot results are preserved rather than recomputed post hoc. These remain descriptive exposure diagnostics because no denominator is yet aligned with the cases. A separate climate- and case-blind terrain pilot enumerates glacier, glacier-contact, and permafrost-intersection objects in three deterministically selected windows. Primary connected-component counts change by factors of 1.3–3.0 between 30 and 90 m grids, so the pilot supplies a reproducible frame but not a unique population count. A post-outcome scale-explicit composite reduces 90 m area departure in five of six paired diagnostics from the same exposed windows. A subsequently registered blind transfer calculation freezes that estimator and tests four hash-selected windows. All eight glacier- and permafrost-stratum checks meet the registered numerical bounds: 90 m departures are 1.8–5.3% and four-phase coefficients of variation are 0.000061–0.00375. The windows are not a probability sample, so this result tests numerical transfer rather than physical susceptibility or global representativeness. A subsequent public, signed-beacon draw selects 96 cells from a frozen 1,826-cell near-global RGI 7-intersecting frame with known geographic inclusion probabilities. Exact source replay then stages four 30 m elevation phases and one PZI subset per cell. A registered coverage-only run stops before any equivalent-area calculation because one selected cell has zero complete DEM-supported reporting centers. Neither complete terrain-screen total is therefore estimable from the frozen source, and no partial estimate is reported. A subsequent full-frame object-inventory snapshot lists the core GLO-30 object for 1,822 of 1,826 cells and the core GLO-90 object for all 1,826; full nine-object halo support is 1,411 and 1,422 cells, respectively. These are asset-metadata counts and do not authorize source substitution or establish valid terrain coverage. A separately frozen RGI-geometry screen then

classifies 1,065 of 1,073 absent GLO-30 cell-object incidences and 1,030 of 1,031 absent GLO-90 cell-object incidences as outside the conservative glacier-proximity screen; the remaining 8 and 1 incidences, in 6 and 1 cells, remain unresolved. This geometric disjointness does not establish DEM computability or terrain validity. I then specify detrended seasonal controls, matched non-failing slopes, mechanism-specific glacier and permafrost observations, and multiplicity control. Uneven observation and unresolved primary sources prevent these counts from estimating a frequency trend. No systematic climatic association or causal effect is estimated here.

1 Introduction

Alpine mass movements become especially hazardous when an initial failure changes material or enters a confined channel. The 2021 Chamoli rock and ice avalanche transformed into a debris flow and flood that scoured valley walls and struck two hydropower projects [21]. Fjord failures instead transfer momentum to water. The September 2023 Dickson Fjord rockslide generated near-source runup of about 200 m and a very-long-period signal that persisted for nine days [18, 23]. The August 2025 Tracy Arm landslide produced a measured maximum runup of 481 ± 7.6 m [22]. A useful catalog must describe the initial failure separately from the downstream cascade.

The mechanical origin of the initial failure remains the central uncertainty. Warming may remove glacier support from a rock slope, thaw ice in fractures, or change meltwater pressure beneath a glacier. Weather may provide a shorter perturbation through rain, snow loading, melt, or rapid temperature change. Lithology, adverse structure, progressive damage, valley geometry, glacier dynamics, and earthquakes remain competing or interacting explanations. A climatic trend can therefore precondition a failure without being its immediate trigger, and a warm event year does not establish causation.

Here, I use a literature seed to define a reproducible catalog and the tests that would be needed to evaluate systematic association with climate. The seed is a protocol-development set, not an inference sample. A separate, climate-blind discovery inventory supplies the candidate sampling frame. The two are kept separate so that examples chosen because they motivated a climate hypothesis cannot enter an association test without being rediscovered under the same rules as every other event.

2 Catalog construction and sampling protocol

The catalog consists of three linked tables. The event table records identity, group, analysis role, date, location, setting, initial failure, and downstream transformations. Dates and coordinates have field-specific sources. The measurement table stores one quantity per row and distinguishes a point value, approximation, lower bound, upper limit, or range. It records uncertainty type, observation period, scenario, evidence kind, and source locator separately. The claim table labels each statement as observation, preliminary observation, reconstruction, model output, published inference, or project interpretation. Each event has explicit process-chain, immediate-trigger, and climate-preconditioning claims, including an unresolved claim when evidence is absent.

The registered geophysical inclusion rule requires rapid failure of rock, glacier ice, or a mixed mass in an alpine or fjord setting, a documented time and source location, and at least one of three operational thresholds: initial volume of 10^6 m³, source-to-deposit travel of 2 km, or landslide-tsunami field runup of 10 m. Downstream damage and population exposure are stored as consequences but do not determine geophysical inclusion. Snow avalanches without glacier-ice or rock failure and floods without an initial mass movement are outside the scope. These thresholds were frozen before the climate-blind screen. A separately labeled active slope may enter the protocol-development set

Table 1: Cause-blind flow through the two enumerated discovery frames. “Retained” applies the occurrence window and a usable event record; the numeric prefilter uses only published volume, travel, or runup fields. It does not establish material or setting eligibility.

Frame	Published	Downloaded	Window	Retained	Numeric
PANGAEA landslide tsunamis	354	355	87	84	40
High Mountain Asia	60	60	40	40	38

when measured deformation and published failure geometries exist. Barry Arm is such a prospective case, not an occurred event [3–5, 20].

Discovery covers 1 January 2000 through 30 August 2026. Two enumerated frames are the 354-case PANGAEA landslide-tsunami catalog [6] and the 60-event High Mountain Asia rock- and ice-avalanche inventory [27, 28]. A 51-event global review supplies a cross-check [17]; its linked spreadsheet was not enumerated because automated retrieval returned HTTP 403. Independent query suites for fjord and lake tsunamis, glacier failures, and rock or mixed avalanches supplement dates and regions not covered by the structured frames. All exact queries, execution times, and resource classes are archived with the catalog. Searches and candidate fields omit asserted climate cause and event climate exposure. Inventory cause columns are ignored.

Every candidate promoted to detailed screening has two differently named reviewers. Each reviewer sees identity, time, place, initial material and process, the threshold observation, and its source. Allowed decisions are include, exclude, or uncertain. An inventory value can discover an event, but final inclusion requires a primary event reconstruction or authoritative agency record. A secondary-only record remains uncertain. Conflicting dates, coordinates, volumes, and aliases remain in the notes rather than being silently reconciled. Detection era, sensor coverage, and reporting source must remain in the later observation model because discovery is not stationary.

The reconciliation crosswalk links all 78 numeric-prefilter rows in Table 1 to candidate records. The source bytes, checksums, and qualifying-row manifest are archived with the catalog. Compound inventory rows are expanded into individual occurrences. A row-level size estimate is not assigned to each component unless the primary source supports that allocation; otherwise each component remains uncertain. Independent searches identified, for example, the November 2025 Bartuytsete ice-rock avalanche, whose published 3.5 km runout passes the travel threshold [7]. The resulting table contains 148 candidates: 73 included, 68 uncertain, and 7 excluded. Fifty-three included failures have day-scale or better timing and enter the future trigger-time risk set; the remainder can inform only site-level or preconditioning analyses.

The seed records were selected because they motivated the study and still cannot support a frequency trend, an event-conditioned climate distribution, or a global susceptibility model. The discovery table is closer to an inference frame, but it also cannot support those estimates until the registered search cutoff, both screens, disagreement resolution, and catalog commit are frozen. No atmospheric anomaly or cryospheric exposure is extracted before that freeze.

The protocol has two estimands. The episode-level trigger estimand compares conditions at failure with time controls; the site-level preconditioning estimand compares source slopes with an incidence-density risk set. The two Aru glaciers are separate episode candidates but share a site-level cluster. Distinct Aysen, Denali, and Kunlun source failures remain individual events even though each set shares an earthquake trigger. The unresolved October 2023 Dickson Fjord episode is linked to the September site but receives no analysis role until its source-specific threshold is established. Barry Arm is excluded from event-time inference because catastrophic failure has not occurred; it instead supplies a prospective case for testing susceptibility variables.

3 Mechanics represented by the seed records

The two Aru glacier tongues detached after surge-like acceleration on low-angle terrain; the published mean Aru-1 surface slope is approximately 13° . Remote sensing and thermomechanical modeling indicate that reduced basal friction shifted resistance toward frozen tongues and lateral margins until those forces were exceeded [12, 16]. Gilbert and colleagues inferred a five- to six-year increase in melt and rain rather than exceptional seasonal forcing. For Aru-1, approximately 10–25 mm of precipitation during the two days before failure is a possible short perturbation; Aru-2 has no resolved immediate trigger. The two collapses traveled approximately 6 and 5 km beyond their respective glacier termini [16].

At Chamoli, $26.9 \times 10^6 \text{ m}^3$ of rock and glacier ice failed from Ronti Peak and transformed into a highly mobile debris flow [21]. The reported 95% reconstruction interval is $26.5\text{--}27.3 \times 10^6 \text{ m}^3$. The mass was about 80% rock and 20% ice; these are model-dependent composition estimates rather than direct material measurements. Mapping and reconstruction do not require a glacial-lake outburst to explain the flood cascade. Geologic structure and progressive damage remain alternatives to thermal and hydrologic preconditioning. Snow and ice loading on the headwall and a preceding positive temperature anomaly have been proposed as final perturbations, but a unique immediate trigger was not isolated [26].

At Dickson Fjord, a foliation-parallel rockslide struck a gully glacier before entering the fjord. The reconstructed rock volume was $25.5 \pm 1.7 \times 10^6 \text{ m}^3$, with a separate $2.3 \pm 0.15 \times 10^6 \text{ m}^3$ entrained-ice estimate [23]. Rockslide, glacier impact, tsunami, and the very-long-period signal are observations or reconstructions. The cross-fjord seiche was first inferred by wave and seismic modeling; sparse SWOT altimetry later provided direct water-level observations consistent with its phase and period [18]. Multidecadal glacier lowering supports a debuttressing hypothesis, but no proximate trigger or formal anthropogenic event attribution was demonstrated.

Tracy Arm and Barry Arm illustrate occurred and prospective consequences of glacier retreat along fjords. The mapped Tracy Arm volume gives a lower limit of $63.5 \times 10^6 \text{ m}^3$ with a reported uncertainty of $2.7 \times 10^6 \text{ m}^3$; unmeasured submarine failure may increase the total. Several days of accelerating microseismicity preceded failure [22]. Regional summer warming since 1875 was quantitatively attributed to anthropogenic forcing. Observed retreat of South Sawyer Glacier supports debuttressing as a mechanical inference, but the glacier retreat itself was not assigned an anthropogenic counterfactual fraction. At Barry Arm, slope displacement and glacier retreat were temporally coincident during 2010–2017 [5]. Their reported correlation of 0.99 compares cumulative trending series and is not by itself a causal test. Published geometry scenarios span $290\text{--}689 \times 10^6 \text{ m}^3$ [3], while a four-layer seismic inversion estimates a local mean failure-surface depth of $188 \pm 9 \text{ m}$ [4]. These are uncertain slope geometries, not a failed volume.

Table 2: Seven focal records in the protocol seed. The dependent October 2023 Dickson Fjord episode remains in the machine-readable catalog but is omitted here. Volumes describe an initial failed mass except at Barry Arm, where the range describes possible unstable-slope geometries. Shown uncertainties are reproduced from the sources; their distinct confidence, conservative-sum, or unspecified semantics are stored in the machine-readable catalog.

Event and date	Initial failure	Volume (10^6 m^3)	Cascade	Climate evidence
Nepal, 26 Aug 2026	Provisional glacial-collapse classification	unresolved	Debris flow and flood traveled about 100 km	No event-specific attribution; process reconstruction remains provisional [24, 25]
Chamoli, 7 Feb 2021	Rock and glacier ice	26.9 [26.5, 27.3]	Avalanche to debris flow and flood	Several thermal effects are plausible; no single trigger isolated [21, 26]
Aru-1, 17 Jul 2016	Glacier detachment	68 ± 2	Ice avalanche; 6 km beyond terminus	Multiyear melt and rain inferred to increase subglacial water pressure [12]
Aru-2, 21 Sep local time 2016	Glacier detachment	83 ± 2	Two pulses; 5 km beyond terminus	Shared multiyear preconditioning; immediate trigger unresolved [12, 16]
Dickson Fjord, 16 Sep 2023	Foliation-parallel rockslide	25.5 ± 1.7 rock	Glacier entrainment, tsunami, and nine-day signal	Glacier thinning inferred to debutstress the toe; proximate trigger unresolved [23]
Tracy Arm, 10 Aug 2025	Bedrock landslide	≥ 63.5 mapped	Glacier/fjord impact, tsunami, and signal lasting at most 36 hours	Regional warming attributed; retreat–debuttressing link is a mechanical inference [22]
Barry Arm, active	Slow bedrock landslide	290–689 possible	Potential rapid failure and tsunami	Retreat and motion are temporally coincident; alternatives require testing [3, 5]

4 Mechanical predictors beyond reanalysis

Atmospheric fields do not measure glacier support, permafrost fracture state, or basal effective pressure. The analysis will therefore pair reanalysis with annual optical and radar mapping of terminus position, glacier-slope contact, surface elevation, and source-slope deformation. Define Δh_i as index-date thickness minus baseline thickness at the slope contact. A signed stress-scale proxy for support loss is

$$S_i = -\rho_i g \Delta h_i, \quad (1)$$

where ρ_i is ice density, so thinning gives positive S_i and thickening gives negative S_i . This proxy has stress units but is not the lateral buttressing traction. Geometry-specific elastic or limit-equilibrium calculations are required to translate it into a support change. Results will therefore report signed thickness change, its magnitude, mapped contact loss, and S_i before interpreting debuttressing.

For a first glacier-detachment screening bound, treat the bed as Coulomb till:

$$\begin{aligned} N &= p_i - p_w, & \tau_y &= c + N \tan \phi, \\ \tau_b &\leq \tau_y, & \tau_d &\simeq \rho_i g h \sin \alpha, \end{aligned} \quad (2)$$

where N is effective pressure, p_i is ice overburden pressure, p_w is water pressure, c and ϕ are effective till cohesion and friction angle, h is ice thickness, and α is surface slope. In a slab that neglects lateral and longitudinal resistance, basal yield cannot balance driving stress when

$$N < N_{\text{crit}} = \frac{\tau_d - c}{\tan \phi}. \quad (3)$$

This is a conditional screening range, not an inferred Aru effective pressure; the analysis must span plausible c , ϕ , geometry, and neglected margin stresses. Melt or rain can raise p_w and reduce resistance when drainage is inefficient. The same input can enlarge efficient drainage, lower p_w , and increase resistance.

Permafrost hypotheses require rock-temperature or fracture-ice observations, not air temperature alone. Where boreholes are absent, the protocol will use modeled rock-surface temperature, topographic shading, fracture orientation, and cumulative thermal history as proxies and will label the inference. Adverse structure is a baseline confounder; deformation may be a mediator or competing-process observation. The causal specification will determine which variables are adjusted rather than treating every measured field as a covariate.

4.1 Climate- and case-blind susceptible-terrain pilot

A denominator for an incidence calculation must be constructed without using the failures it will later explain. I therefore registered a terrain pilot before reading source-product values. It fixes RGI 7.0 regions 01, 11, and 13 and chooses one 1° window per region using only catalog fields [19]. The selector uses the archived `cenlon` and `cenlat` points, which the RGI metadata describes as approximately central points within the outlines rather than geometric centroids. Eligible cells contain at least 10 such points and 1 km^2 of summed glacier area; the cell with the smallest SHA-256 digest of its region and integer edges is selected. This rule gives 58–59°N, 155–154°W in Alaska; 45–46°N, 7–8°E in Central Europe; and 43–44°N, 94–95°E in Central Asia. These are validation windows, not regional samples.

RGI polygons define glacier objects and the ice mask. Native Copernicus DEM GLO-30 values are warped to a local Lambert azimuthal equal-area grid in each window [10]. Centered differences define surface slope at 30 m; exact 3×3 elevation means give the aligned 90 m sensitivity. Outside-glacier

terrain is joined by four-neighbor connectivity. The glacier-contact stratum requires slope of at least 30° and raster edge distance no greater than 100 m. The permafrost stratum applies the same slope threshold and intersects $PZI \geq 0.1$ from the 30-arc-second global zonation map [13, 14]. PZI is a climatological index based on 1961–1990 air temperature, not a measured local temperature or failure probability. The Global PZI data source is the World Glacier Monitoring Service (WGMS), with Gruber as creator and UZH as publisher. The machine-readable tables retain slope thresholds of 25° , 30° , and 35° , contact distances of 0, 100, and 300 m, PZI thresholds of 0.1 and 0.5, and both grid spacings. ITS_LIVE version 2 contributes only its 2014–2022 image-pair count as an observability field; no velocity or trend enters object membership [11].

Counts below include boundary-truncated objects; the archived summary also reports wholly contained objects separately. The organisations in charge of the Copernicus programme by law or by delegation do not incur any liability for any use of the Copernicus WorldDEM-30.

For every object of area A , the calculation reports $V_d = Ad$ for prescribed depths $d = 10, 30$, and 100 m. An object is conditionally eligible when $V_d \geq 10^6 \text{ m}^3$. This does not estimate glacier thickness, rock-failure depth, or mobilized volume. The primary 30 m calculation contains 140, 472, and 144 glacier polygons in Alaska, Central Europe, and Central Asia, respectively. The corresponding glacier-contact counts are 1,766, 2,872, and 742; permafrost-intersection counts are 3,672, 5,321, and 7,099 (Figure 1). At $d = 30$ m, conditional counts are 137, 340, and 117 for glaciers; 331, 611, and 220 for glacier-contact terrain; and 724, 753, and 1,086 for permafrost-intersection terrain. The three strata overlap and these values must not be summed.

Area is less sensitive than object count. From 30 to 90 m, glacier-contact area changes by -8% , $+6\%$, and $+3\%$ in the three regions, whereas its component count decreases by factors of 1.7, 1.6, and 1.3. Permafrost-intersection area decreases by 18%, 11%, and 35%, while component count decreases by factors of 1.9, 2.0, and 3.0. Conditional counts need not vary monotonically: coarsening removes steep pixels but also joins patches. Increasing contact distance similarly increases total area while it can reduce the number of volume-eligible components by merging them. A deterministic audit of five objects per stratum and window reproduced every source-mask predicate, area, connectivity, boundary flag, and coverage field.

4.2 Held-out susceptible-area convergence

The object-count sensitivity above motivates a separate test of area as a possible exposure measure. Before reading new inventory values, I ranked 14 RGI region codes by SHA-256 and fixed regions 07, 08, and 03. A second blind hash selected $77\text{--}78^\circ\text{N}$, $18\text{--}19^\circ\text{E}$ in Svalbard; $69\text{--}70^\circ\text{N}$, $21\text{--}22^\circ\text{E}$ in Scandinavia; and $81\text{--}82^\circ\text{N}$, $85\text{--}84^\circ\text{W}$ in Arctic Canada North. The 200 eligible window keys and their digests were frozen before a DEM or PZI value was read. These deterministic windows are held out from the pilot, but they are not a random sample of mountain terrain.

The registered reference is total area under the pilot’s 30° slope, 100 m glacier-contact, and $PZI \geq 0.1$ screens. I independently warp the DEM and masks to four 30 m grids separated by half-cell phase shifts. The 90 m variant averages complete 3×3 elevation blocks but samples masks on its own grid. A stratum passes only if every window has a 90 m area departure no greater than 20% and a population coefficient of variation no greater than 0.10 across the four 30 m phases.

Grid phase has little effect (Figure 2). The largest phase coefficient of variation is 0.00375. In contrast, coarsening reduces glacier-contact area by 26.7% in Arctic Canada and 36.6% in Svalbard; the change is 0.7% in Scandinavia. PZI-intersection area decreases by 29.1%, 37.4%, and 14.2%, respectively. Both strata therefore fail the registered all-window criterion. This comparison suggests that half-cell alignment is not the controlling numerical uncertainty among the tested variants. Resolution coarsening changes DEM smoothing, slope support, mask sampling, contact-distance

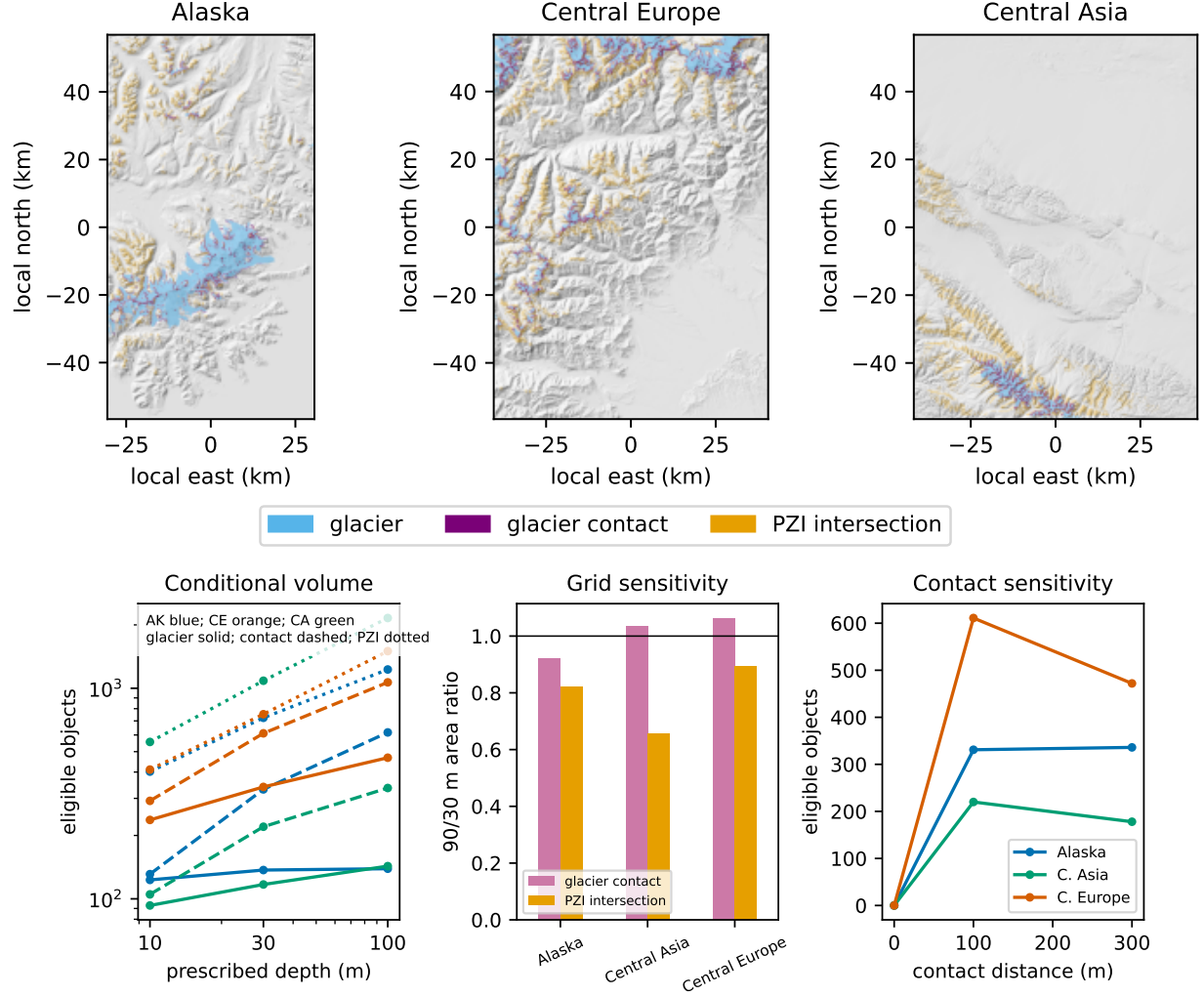


Figure 1: Climate- and case-blind susceptible-terrain pilot. Top: the three deterministically selected windows with RGI glacier polygons, primary steep glacier-contact terrain, and primary steep PZI-intersection terrain. Bottom left: conditional-volume counts across prescribed depths. Bottom centre: primary rock-slope 90/30 m area ratios. Bottom right: conditional counts across glacier-contact distances. Counts include boundary-truncated objects. In the depth panel, colors distinguish regions and line styles distinguish strata. Strata overlap, source epochs differ, and none of the mapped objects is a predicted failure. Elevation products were produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA; all rights reserved.

discretization, reporting centers, and cell area together, so this test does not isolate which operation causes the departure. Neither area measure is used below to infer a global incidence density or hazard distribution.

The registered geometric choices also remain influential. Across the three windows, the unshifted-reference-grid 30 m sensitivity calculations span 0.27–114.0 km² for glacier-contact area and 13.6–403.4 km² for PZI-intersection area as slope, contact distance, and PZI threshold vary. These ranges describe the screens; they are not uncertainty intervals or alternative opportunities to pass the convergence test.

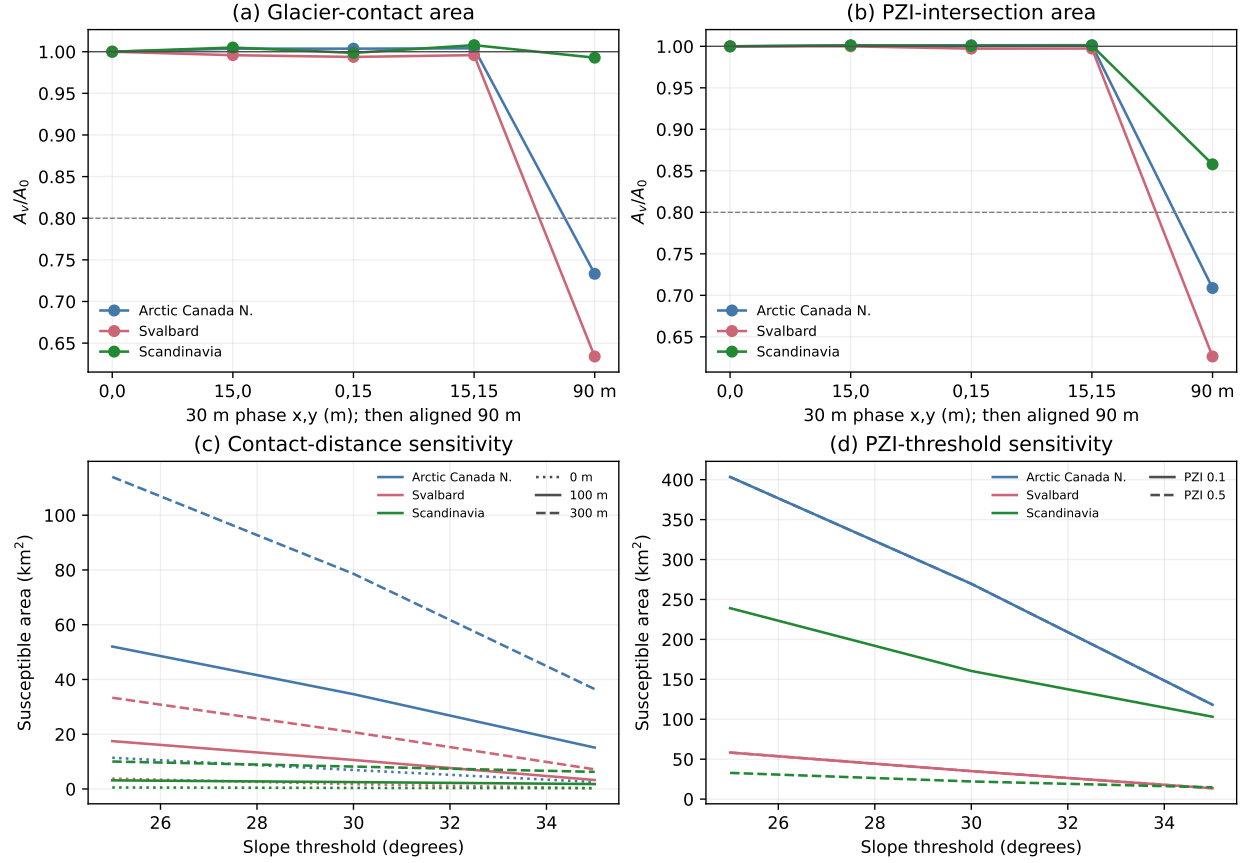


Figure 2: Held-out susceptible-area convergence. Panels (a) and (b) show area relative to the unshifted 30 m result for three half-cell phase grids and the aligned 90 m grid. The dashed 0.8 line is the lower registered resolution bound and applies to the 90 m point; phase is assessed by its four-grid coefficient of variation. Panels (c) and (d) show separate 30 m sensitivity to slope, contact distance, and PZI threshold. Colors distinguish held-out regions and line styles distinguish geometric thresholds. The plotted objects are not predicted failures. Global PZI data are attributed to WGMS as source, Gruber as creator, and UZH as publisher. Elevation products were produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA; all rights reserved.

4.3 Exposed-window scale-explicit development

The failed hard-threshold area test motivates a smoother numerical representation, but its values also exposed the windows used to develop that representation. I therefore registered the following calculation as post-outcome method development rather than a second holdout. It reuses the same DEM phases, glacier inventory, PZI samples, reporting windows, and local equal-area projections. No event, climate, deformation, or damage field enters the calculation.

At each target-cell center, I fit a vertical-residual plane to all finite DEM centers within a nominal 300 m radius. For center offsets (u, v) on a grid of spacing r , symmetry gives

$$a = \frac{\sum uz}{\sum u^2}, \quad b = \frac{\sum vz}{\sum v^2}, \quad q = (a^2 + b^2)^{1/2}. \quad (4)$$

The steepness weight is zero below $q = \tan 25^\circ$, one above $q = \tan 35^\circ$, and linear in q between those values. The glacier-outline stratum contains target centers outside the inventory union whose exact planar distance to it is no greater than 100 m. The permafrost stratum contains outside-glacier centers with PZI at least 0.1. Equivalent steep area is $E = r^2 \sum w$ over reporting centers in each stratum with complete plane support. Although E has area units, it is a weighted numerical measure rather than literal susceptible area or failure probability.

The four 30 m phases remain similar: population coefficients of variation range from 0.00012 to 0.00347 (Figure 3). Relative to the unshifted 30 m value, 90 m departures for glacier-outline proximity and PZI intersection are 5.9% and 3.7% in Arctic Canada North, 13.4% and 7.1% in Svalbard, and 3.7% and 2.6% in Scandinavia. These are smaller than the hard-threshold departures in five of six paired diagnostics. The exception is Scandinavian glacier proximity, which increases from 0.7% to 3.7%. All six new departures lie below the former 20% bound, but that line is a visual reference and does not create a confirmatory pass result in exposed windows.

Fixed support, continuous weighting, and exact outline proximity change together. Their paired difference therefore belongs to the composite and cannot identify which operation controls scale dependence. Moreover, the 30 and 90 m disks contain different center quadrature; the calculation is not a resolution-independent physical kernel, failure-plane orientation, thickness, or stress. The blind-region calculation below therefore freezes this estimator unchanged before reading additional terrain values. These exposed-window diagnostics do not themselves support global scaling.

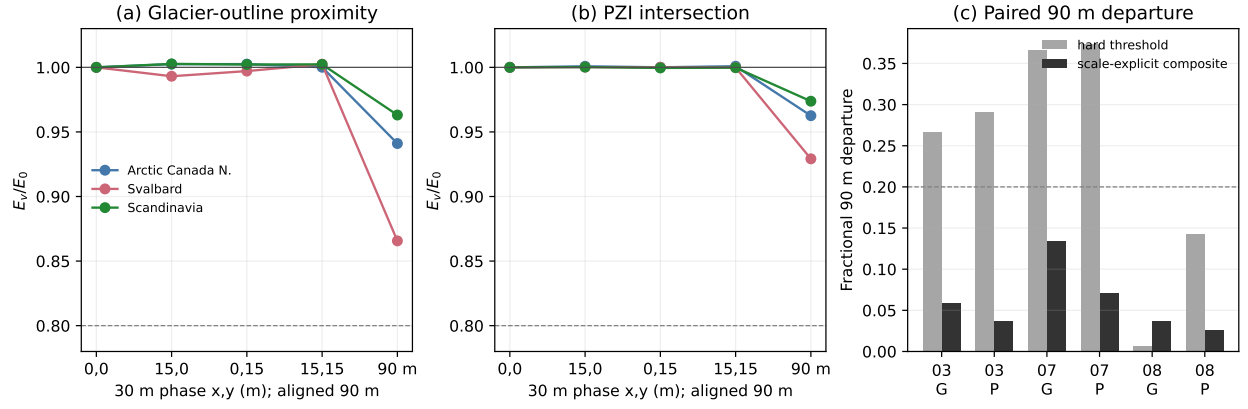


Figure 3: Known-window scale-explicit equivalent-area development. (a, b) Equivalent steep area relative to the unshifted 30 m result for four 30 m phases and the aligned 90 m grid. (c) Paired 90 m departures for the failed hard-threshold screen and the new composite; G denotes glacier-outline proximity and P denotes PZI intersection. The dashed 0.8 ratio and 0.20 departure lines repeat former bounds only as references. The calculation was selected after the hard-threshold results were known, has no pass label, and contains no event or predicted-failure marker. Source products and attribution are as in Figure 2.

4.4 Blind numerical transfer of equivalent steep area

The transfer calculation tests whether the frozen equivalent-area estimator retains its numerical behavior outside the three development windows. Before terrain values were opened, a registered protocol excluded previously used regions, enumerated inventory-only eligible windows, and selected four RGI regions and one 1° window per region by SHA-256 order. The resulting windows are Arctic Canada South ($72\text{--}73^\circ\text{N}$, $79\text{--}78^\circ\text{W}$), North Asia ($65\text{--}66^\circ\text{N}$, $145\text{--}146^\circ\text{E}$), South Asia East ($28\text{--}29^\circ\text{N}$, $96\text{--}97^\circ\text{E}$), and New Zealand ($44\text{--}43^\circ\text{S}$, $171\text{--}172^\circ\text{E}$). Inventory attributes fixed the selection; glacier geometry, elevation, and PZI values were acquired only after the selection manifest was frozen.

The protocol retained the development calculation without changing the 300 m support radius, $25\text{--}35^\circ$ ramp, 100 m glacier-outline distance, PZI threshold of 0.1, or grid phases. For region j and stratum s , let $E_{js,0} = E_{js,00}$ and $E_{js,90}$ denote the unshifted 30 m and aligned 90 m values. The four 30 m values $E_{js,00}$, $E_{js,10}$, $E_{js,01}$, and $E_{js,11}$ have respective phase offsets (0, 0), (15, 0), (0, 15), and (15, 15) m. The protocol defined the departure and population coefficient of variation as

$$D_{js} = \frac{|E_{js,90} - E_{js,0}|}{E_{js,0}}, \quad C_{js} = \frac{\text{sd}_{\text{pop}}(E_{js,00}, E_{js,10}, E_{js,01}, E_{js,11})}{\text{mean}(E_{js,00}, E_{js,10}, E_{js,01}, E_{js,11})}. \quad (5)$$

Transfer required a positive finite $E_{js,0}$, $D_{js} \leq 0.20$, and $C_{js} \leq 0.10$ in both strata of all four windows. Thus, one failed window-stratum check would fail its stratum and the complete test.

Source replay reproduced all 20 derived DEM and PZI grids with matching value hashes, equal affine transforms, and zero maximum absolute difference. The first computation stopped when two WGS84-valid glacier outlines became invalid after projection in South Asia East. Regions 10 and 18 had been calculated in memory, but no result was persisted, printed, summarized, or observed. Before restarting, the pre-output provenance recorded a deterministic linework repair applied only to invalid projected geometry; the two repaired areas changed by less than 8.3×10^{-16} in relative magnitude. This repair addresses polygon representation after projection and does not alter the frozen steepness or comparison equations.

All eight reference areas are positive and span $0.52\text{--}297.42 \text{ km}^2$. Figure 4 shows that all eight checks pass. The 90 m departures range from 1.81% to 5.28%, compared with the registered 20% bound. The largest is the New Zealand PZI-intersection result. The four-phase coefficients of variation range from 0.000061 to 0.00375, compared with the registered 0.10 bound; the largest is the Arctic Canada South glacier-outline result. The complete result therefore passes the registered numerical transfer gate.

This pass has a narrower interpretation than model validation. It confirms resolution and phase behavior for the frozen calculation in four additional hash-selected windows. It does not show that the two strata identify unstable slopes, resemble event sources, predict incidence, or sample the global terrain distribution. Neither the four RGI regions nor their first eligible hash windows have known geographic inclusion probabilities. A separate sampling protocol is therefore required before estimating a regional or global equivalent-area denominator.

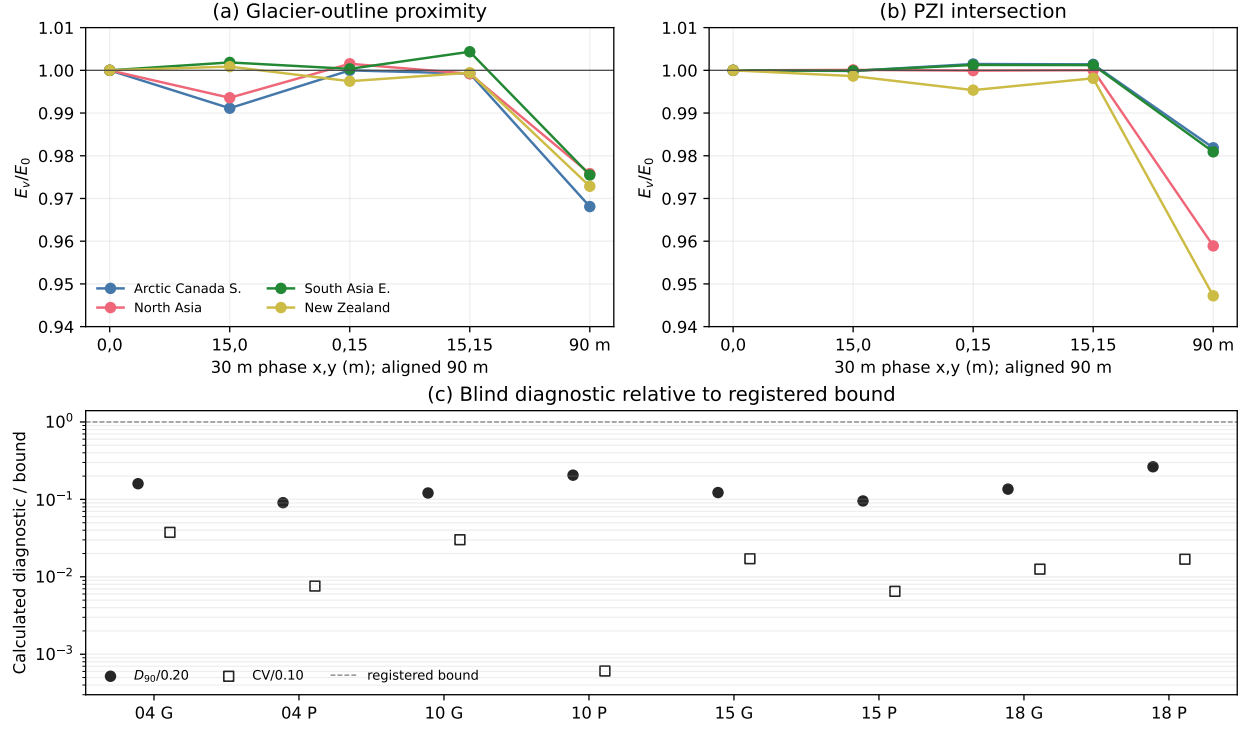


Figure 4: Blind numerical transfer of scale-explicit equivalent steep area. (a, b) Equivalent area relative to the unshifted 30 m result for four 30 m phases and the aligned 90 m grid in glacier-outline proximity and PZI intersection strata. Colors identify four regions selected by the frozen inventory-only hash procedure. (c) Each 90 m departure and phase coefficient of variation divided by its registered bound; G denotes glacier-outline proximity and P denotes PZI intersection. The dashed line is the acceptance threshold. All values lie below it. This is a numerical transfer result, not a physical-validation or global-sampling result. Source products and attribution are as in Figure 2.

4.5 Geographic probability sample

The numerical transfer result motivates a finite-population estimate, but its four windows cannot supply one. Before opening any new DEM or PZI value, a registered design intersected unique RGI 7 glacier outlines with global 1° reporting cells. It retained cells from 60°S through 89°N having positive glacier-intersection area. Where outlines from several RGI regions entered one cell, the region having the greatest geodesic union area defined its sampling stratum. This produced 1,826 cells in 19 nonempty regional strata. RGI reports zero glaciers in region 20; accordingly, no archive is distributed and its frame population is zero. Independent reconstruction from all 19 source archives reproduced 274,531 unique features, the complete frame, its regional ownership, and its byte hash.

The fixed total sample size was 96. Each non-census stratum first received four cells, and remaining cells were assigned by the registered quotient rule. The realized allocations are 4–12 cells per nonempty region. After the exact frame commit was posted publicly, the protocol imposed a one-hour delay and requested the first NIST Randomness Beacon 2.0 pulse after the resulting target. The signed pulse occurred 34 s after that target. Its certificate identifier, certificate path and validity, current and preceding RSA signatures, output hashes, pulse link, 60-s index step, and precommitment all verify independently. The verified 512-bit output keyed the frozen HMAC ordering; it did not change the frame or allocations.

Figure 5 shows the complete draw. All 1,826 cell digests are unique, and each regional selected set is the registered prefix. Thus every cell has first-order probability n_h/N_h , and two distinct cells in the same stratum have joint probability $n_h(n_h - 1)/[N_h(N_h - 1)]$; probabilities multiply across strata. A retained chance-balance diagnostic shows that none of the 56 frame cells intersected by more than one RGI region appears in the sample. Under the registered stratified design, the probability of zero is 0.0121. This is not a replacement or redraw criterion, but it prevents interpreting the realized sample as observed cross-region coverage.

This draw supplies known inclusion probabilities for Horvitz–Thompson totals of glacier-outline-proximity and PZI-intersection equivalent steep area. The two masks can overlap and remain separate outcomes. After independent pre-outcome approval, source staging retained 693 available Copernicus DEM objects and 83 registered HTTP 404 responses. It produced 384 phase-specific DEM rasters and 96 PZI subsets. All 480 staged arrays replay from retained source bytes with identical transforms and value hashes and zero maximum absolute difference.

The coverage-only calculation stopped before evaluating equivalent area. Figure 6 isolates the failed p00 cell, whose south and west edges are 41°N and 48°E . All six DEM objects covering the reporting-cell latitude and southern halo returned HTTP 404; only three objects in the northern halo row were available. Thus none of 10,303,590 reporting centers had complete DEM support. In contrast, the RGI predicate was computable at all reporting centers and PZI was finite at all 10,302,530 outside-RGI centers. The registered rule required positive complete DEM support in every selected cell and phase. It therefore stopped before atomic coverage writes and before any equivalent-area calculation.

The failed cell belongs to a stratum with $N_h = 29$ and $n_h = 4$. Its registered outcome is undefined rather than zero, so neither stratum mean nor either complete 1,826-cell total can be formed. No unit was redrawn, deleted, imputed, or assigned a substitute source, and no partial total is reported. This is source-specific non-estimability, not evidence of zero equivalent terrain area or low physical susceptibility.

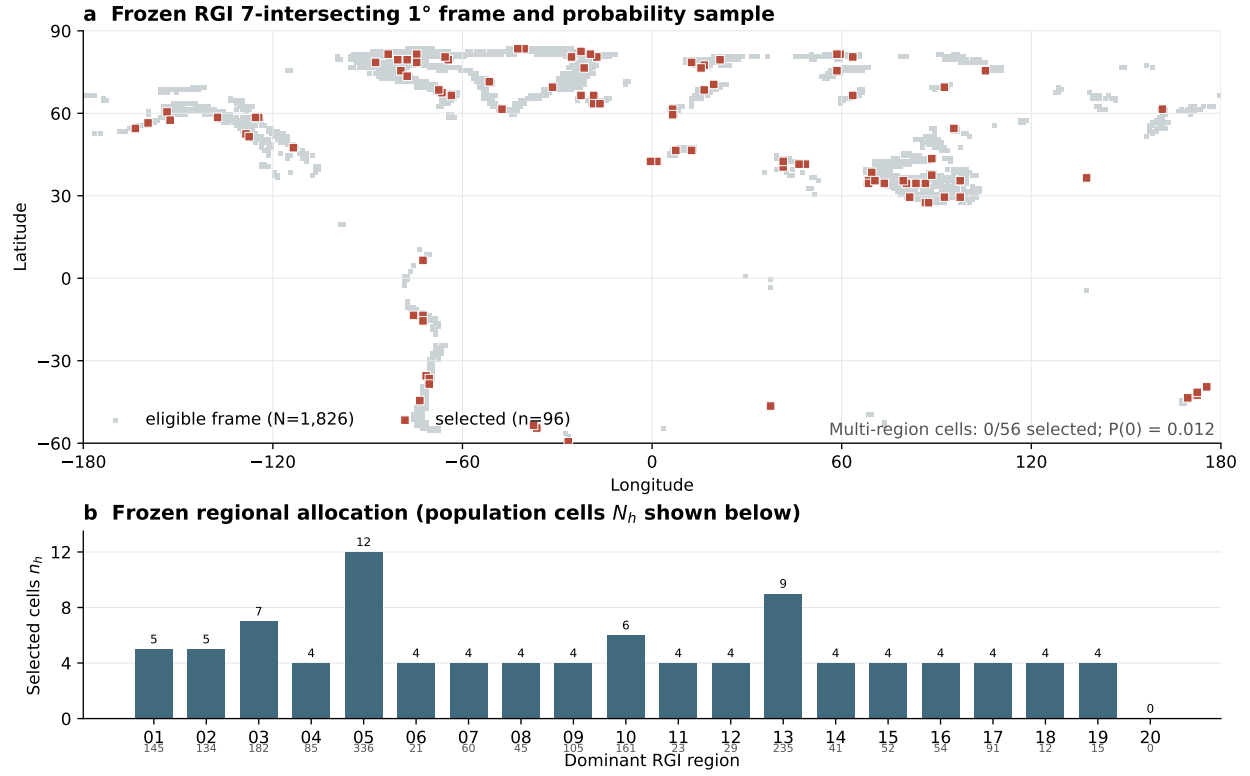


Figure 5: Frozen geographic probability frame and signed-beacon draw. (a) Centers of all 1,826 eligible 1° cells and the 96 selected cells. The probability of selecting none of the 56 multi-region cells is a post-randomization balance diagnostic under the frozen design; no redraw was made. (b) Selected count in each of 19 nonempty dominant-region strata; population counts appear below the region codes. Empty RGI region 20 is shown with $N_{20} = n_{20} = 0$. The figure describes sampling support only; no DEM, PZI, climate, event, or terrain-outcome value entered the frame or draw.

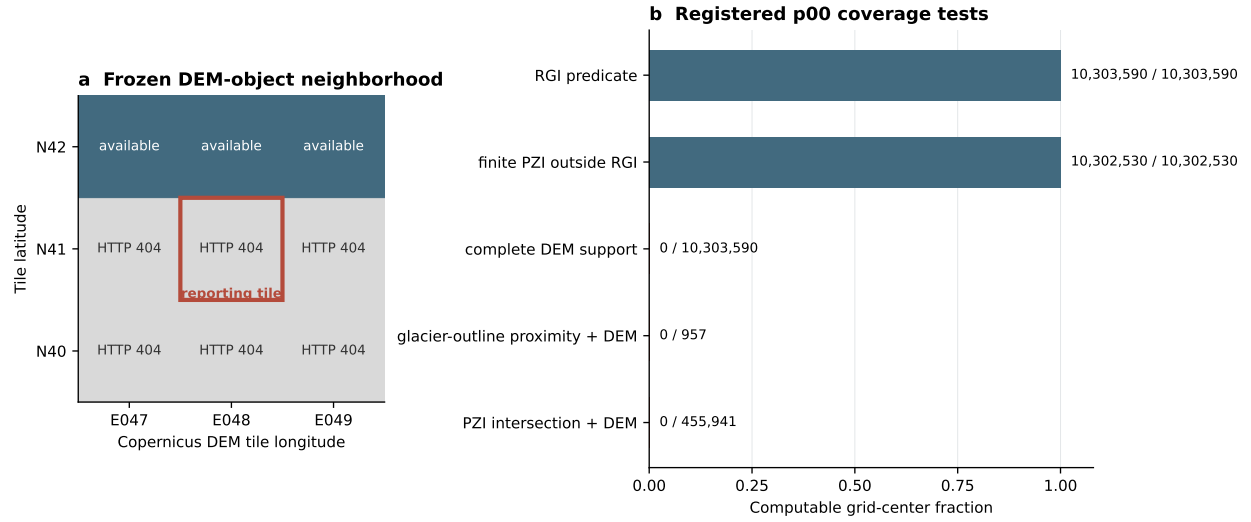


Figure 6: Registered source-coverage stop before geographic terrain inference. (a) Availability of the nine Copernicus DEM objects required by the failed p00 processing grid. The outlined N41–E048 object contains the 1° reporting cell. All six objects in the reporting latitude and southern halo returned HTTP 404; only the three northern-halo objects were available. (b) Computable grid-center fractions under the frozen predicates. The RGI predicate and finite PZI outside RGI have complete coverage at their respective denominators, whereas complete DEM support is zero. The two lower rows restrict the same DEM test to glacier-outline-proximity and PZI-intersection centers. These are source-computability fractions, not susceptible-area fractions. The zero DEM-support fraction triggered the registered stop; no equivalent area or design-based total was calculated.

4.6 Full-frame DEM object-support audit

The selected-cell failure could reflect a local source gap or a broader frame property. A subsequent protocol therefore enumerated the complete public Copernicus DEM GLO-30 and GLO-90 object inventories before opening any new raster. The fixed population remained all 1,826 reporting cells, and each cell required its core 1° object plus the eight neighboring objects needed by the registered 1 km processing halo. The 96 selected identities were retained only as annotations and did not select a request, object, or summary. The anonymous bucket listings were acquired on 1 September 2026 and retained as 848 response pages with 242,695,411 source-metadata bytes [1, 2].

Figure 7 shows the finite-population counts. GLO-30 lists the core object for 1,822 of 1,826 cells (99.78%), whereas GLO-90 lists all 1,826 core objects. The four GLO-30 absences are N39E044 and N41E046, N41E047, and N41E048; all belong to dominant RGI region 12. The last is the reporting object in the selected cell that stopped the registered calculation. Full nine-object halo support is lower: 1,411 cells (77.27%) for GLO-30 and 1,422 (77.88%) for GLO-90. Of 4,240 distinct required spatial identities, 544 are absent from both inventories and 22 additional identities are absent only from GLO-30. Thus 1,411 cells have full-halo support in both products, 404 lack it in both, and 11 lack it only in GLO-30. These are exact enumeration counts and therefore have no sampling standard error.

Listed-object metadata is necessary but not sufficient for terrain calculation. It does not establish that a raster is readable, that its values are valid over snow and ice, or that its acquisition epoch is suitable. In particular, an omitted ocean-only neighbor cannot be assumed irrelevant to a PZI-intersection center without a separately frozen mask-support calculation. The object audit therefore does not revise the Issue 19 stop, substitute GLO-90, construct a mixed mosaic, or estimate equivalent area or physical susceptibility.

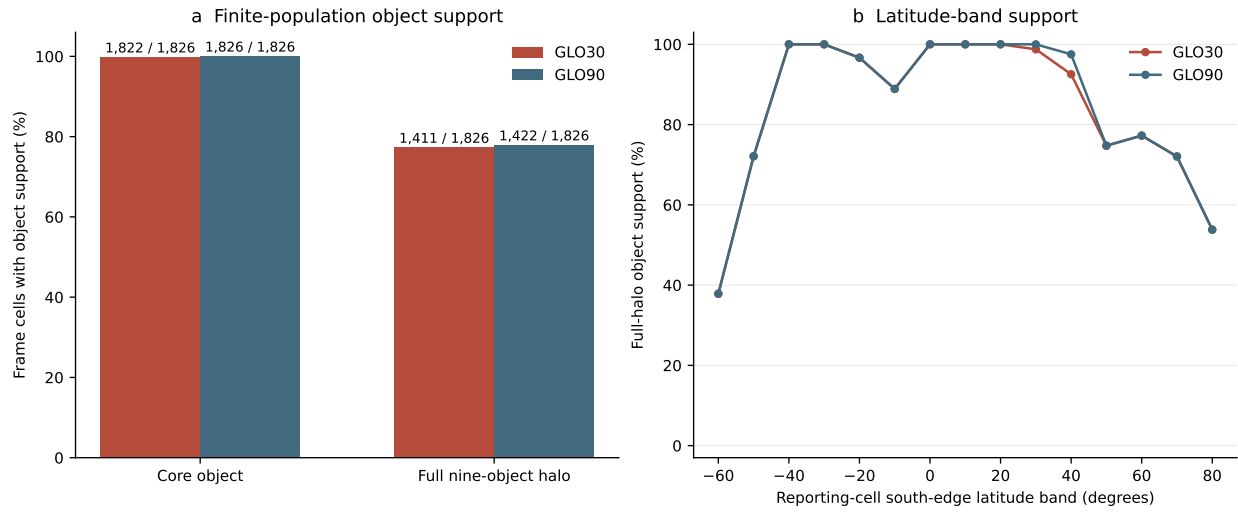


Figure 7: Full-frame Copernicus DEM object-inventory support on 1 September 2026. (a) Fraction and exact count of the 1,826 reporting cells having the listed core object or all nine listed processing-neighborhood objects. GLO-90 lists every core object; four GLO-30 core objects are absent. (b) Full-halo object-support fraction grouped by the 10° band of the reporting-cell south edge. The two products differ only in the 30° and 40° bands; lower shared fractions reflect absent neighborhood objects, including ocean-only tiles. These are object-metadata fractions, not terrain-validity, equivalent-area, susceptibility, or hazard fractions.

4.7 Glacier-proximity screen for absent DEM cell-object incidences

The full-halo audit treats every absent object as potentially consequential, including ocean-only neighbors. A separately frozen calculation therefore asked the narrower geometric question of whether each nominal delivery unit was disjoint from a conservative screen around the registered glacier-outline-proximity continuum. For every one of the 1,826 frame cells, the calculation screened the nine fixed spatial object footprints against all complete RGI 7 outlines. It used a 101 m outward glacier buffer and a closed 1,001 m footprint-distance predicate, deliberately broader than the registered 100 and 1,000 m distances. The 16,434 spatial comparisons were sealed before joining either product inventory, then expanded to 32,868 product incidences. This numerical conservativeness is relative to the registered 0.001° computational outline, not physical outline uncertainty or an exact curved coordinate transformation. Projection validity repairs were retained in a source-outline-identified ledger and passed absolute area-change, relative area-change, and boundary displacement bounds.

Figure 8 reports the exact finite-population result. Among the 16,434 GLO-30 cell-object incidences, 15,361 are listed, 1,065 are absent but outside the conservative screen, and 8 absent incidences intersect the screen and remain unresolved. The corresponding GLO-90 counts are 15,403, 1,030, and 1. At the cell level, 1,820 GLO-30 cells and 1,825 GLO-90 cells have no absent object intersecting the screen; 6 and 1 cells remain unresolved, and none is not applicable. All four absent GLO-30 core cell-object incidences intersect the conservative screen, so this screen does not alter the regional gap that stopped the registered calculation. The unresolved-cell fractions are concentrated in three northern 10° bands for GLO-30 and one for GLO-90, but these are enumeration summaries rather than sampled estimates and have no sampling standard error.

The screen supports only a geometric exclusion: a disjoint absent cell-object incidence lies outside the registered conservative glacier-proximity screen. Intersection remains unresolved. Neither label establishes object readability, finite elevation values, terrain validity, or exact source dependence. The result therefore does not revise the Issue 19 stop, substitute GLO-90, construct a mixed mosaic, or estimate equivalent area, susceptibility, hazard, or climate attribution.

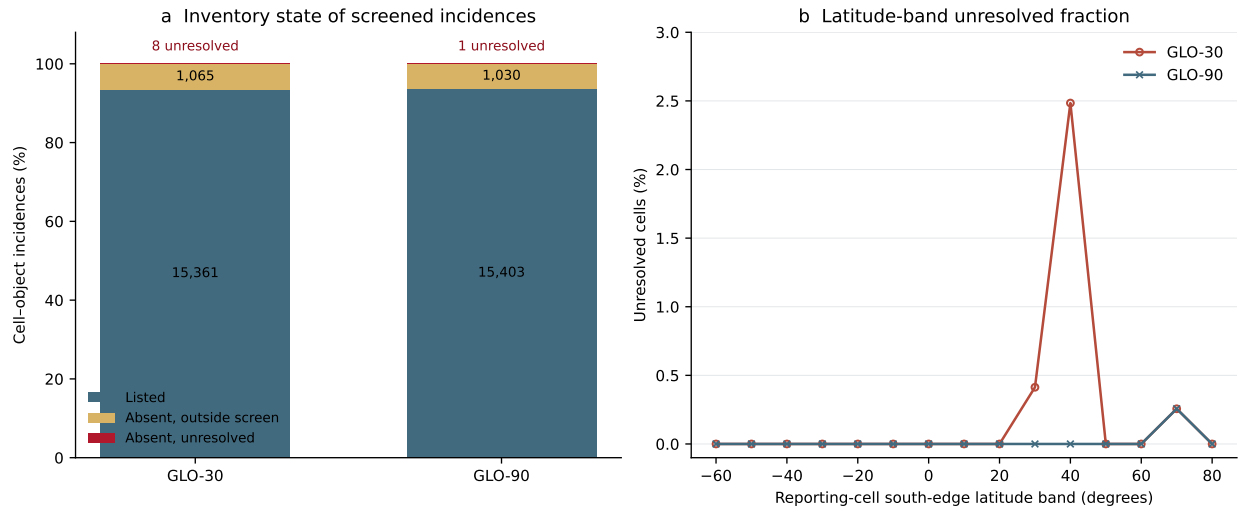


Figure 8: Conservative glacier-proximity screen for absent Copernicus DEM cell-object incidences. (a) Exact inventory-state counts among the 16,434 fixed incidences for each product. The thin red segments contain eight unresolved GLO-30 incidences and one unresolved GLO-90 incidence. (b) Exact unresolved-cell percentage by the 10° band of the reporting-cell south edge. Zero bands are retained. Disjointness places an absent cell-object incidence outside the conservative geometric screen; intersection remains unresolved. Neither state establishes raster readability, finite elevation support, terrain validity, or exact source dependence.

5 Event-centered climate test

For each frozen event candidate, reanalysis will provide near-surface air temperature, precipitation phase and amount, snow accumulation, radiative and turbulent fluxes, and surface pressure. Prespecified reductions will include cumulative positive degree days, liquid-water input, recent snow load, freeze–thaw crossings, and a multidecadal temperature trend. Windows, directions, and mechanism-specific primary contrasts will be registered before event values are examined.

Immediate-trigger anomalies and long-term preconditioning require different controls. For trigger variables, I will remove the local seasonal cycle and a smooth long-term trend using data outside the event window. The event residual will be compared with time-matched, same-season blocks. Block resampling will preserve serial correlation, and the two Aru records will share a site-level cluster. This avoids a null that ranks recent event years against an earlier, cooler climate and then interprets the resulting rank as unusual weather.

For persistent preconditioning, the comparison will use incidence-density risk sets. At each event date, eligible controls are catalog-search source slopes that remain under observation and have not yet failed. Controls receive the case index date, and every exposure window ends before that date; a control may later become a case. Follow-up ends at failure, loss of adequate sensor coverage, or the registered study end. Matching uses baseline region, elevation, aspect, relief, lithology where available, and pre-exposure sensor coverage. It does not use index-date glacier contact or deformation, which may lie on the causal pathway. A preregistered causal diagram will distinguish baseline confounders from exposures and mediators.

A hierarchical model will estimate associations while including region and site-group effects, detection era, and monitoring intensity. Glacier-detachment sites will have a signed liquid-water/effective-pressure contrast; glacier-contact rock slopes will have a signed contact-loss/support contrast; and permafrost rock slopes will have a cumulative thermal contrast. Predictors will be standardized against their own risk-set distributions and will not be imputed across inapplicable mechanism classes. Family-wise error will be controlled across these primary contrasts; secondary predictors will use a false-discovery-rate procedure. An association under this design would still not constitute a complete anthropogenic event attribution.

Global products smooth the elevation and aspect that govern steep-terrain energy balance. Downscaling will begin with corrections tied to a named bias: temperature adjusted between grid-cell and source elevation using a documented lapse rate, precipitation separated by phase, and shortwave radiation corrected for slope, aspect, and shading. Results will be reported before and after each correction. Regional modeling is warranted only when these corrections change an event anomaly enough to affect the prespecified test.

5.1 Source-coordinate and UTC-onset audit

After the registered temperature pilot exposed location and clock ambiguity, I registered a separate, climate-blind audit before retrieving any new event exposure. It fixes the same 53 trigger-time-eligible occurrences and preserves every original catalog value. The coordinate target is the initiating source or source-area centroid, not a named summit, deposit, lake, gauge, dam, or downstream impact. Each assertion records geometry role, evidence method (published, digitized, gazetteer, or unavailable), a horizontal-uncertainty class that may be unbounded, exact source locator, and reviewer state. Digitized points are accepted only when map control supports the registered uncertainty class.

Time assertions target the earliest documented motion of the qualifying source failure and are stored as half-open UTC intervals. Local clocks without a documented civil scale are not converted from longitude. Multi-pulse events retain the qualifying pulse range when source volume cannot

be allocated by pulse. An earthquake origin can be retained only after an event-specific source establishes that the named failure was coseismic; it remains labeled a trigger-origin proxy rather than a measured detachment clock. Two differently named reviewers check each assertion. The final tables, validator, protocol, and retrieved legally redistributable source objects are checksummed in a manifest frozen before any new reanalysis extraction. Seven otherwise reusable objects could not be archived because automated retrieval was blocked; their access failures remain explicit in the source table.

The audit accepts source coordinates for 24 of 53 occurrences: 21 have uncertainty bounded at 1 km and 3 at 5 km. It accepts UTC intervals for 34, comprising 13 minute, 11 second, and 10 range assertions; 3 more retain conflicting source intervals and 16 remain unresolved. Nine accepted intervals are earthquake trigger-origin proxies, not measured detachment times. Twenty-two occurrences have both an accepted source coordinate and UTC interval (Figure 9). Completeness is strongly source-dependent: accepted-coordinate counts include 0 of 7 High Mountain Asia frame records, 13 of 20 independent rock-search records, and 2 of 2 web-search records. The small cells and evidence-availability differences do not describe occurrence rates.

Among the 29 occurrences in the registered ERA5 pilot, only 9 have accepted source coordinates, 17 have accepted UTC intervals, and 9 have both; 2 retain time conflicts. For these 9 occurrences, the largest catalog-to-audited separation is 0.677 km and the median is zero. That small conditional distance does not validate the other 20 catalog points: their main limitation is target geometry or unbounded uncertainty rather than decimal precision. Five Denali source-scar digitizations were accepted only after independent map review. Conversely, EarthScope timestamps for Elliot Creek and Pedersen were retained as station arrivals rather than source onset. An independently reviewed USGS second supports Pedersen; Elliot remains unresolved, and two Paat primary sources remain separated by two seconds. The audit thus changes which observations qualify, even where accepted numeric coordinates move little.

5.2 Registered ERA5 temperature pilot

After freezing the discovery inventory, I registered a narrower extraction test before reading temperature at an event location. It selects every included, trigger-time-eligible occurrence with finite catalog coordinates, giving 29 occurrences. These are the original discovery coordinates, not the later accepted source-coordinate set. Hourly 2-m temperature comes from the public temporal-layout Icechunk copy of ERA5 snapshot T9H8SG2PVXWNYOQNJPJG [8, 15]. Surface geopotential and land fraction come from the NSF NCAR 0.25-degree ERA5 invariants [9]. Three fixed, non-event probes matched the temporal and spatial layouts exactly before event extraction.

The finite-catalog-coordinate requirement retains 29 of the 53 trigger-time-eligible occurrences, and this input availability is strongly source-dependent: coordinates are present for 2 of 20 records from the independent rock-search suite, 11 of 12 from the landslide-tsunami frame, and all 7 from the High Mountain Asia frame. The later audit accepts source geometry for only 9 of these 29. The summaries below therefore characterize the registered catalog-point subset, not all 53 eligible occurrences or a source-verified subset.

For each occurrence, the extraction retains the two nearest distinct grid coordinates on each axis. The primary cell has the largest land fraction, with distance and coordinates resolving ties; results also retain all four cells. The registered primary window contains the 168 complete UTC hours before 00:00 UTC on the published catalog date. The source audit shows that only 17 of the 29 occurrences have accepted UTC intervals, while 2 conflict and 10 remain unresolved. In particular, the Aru-2 local date makes the original window include three post-onset hours. I therefore retain the registered pre-date result and add a conservative sensitivity covering the seven days ending one

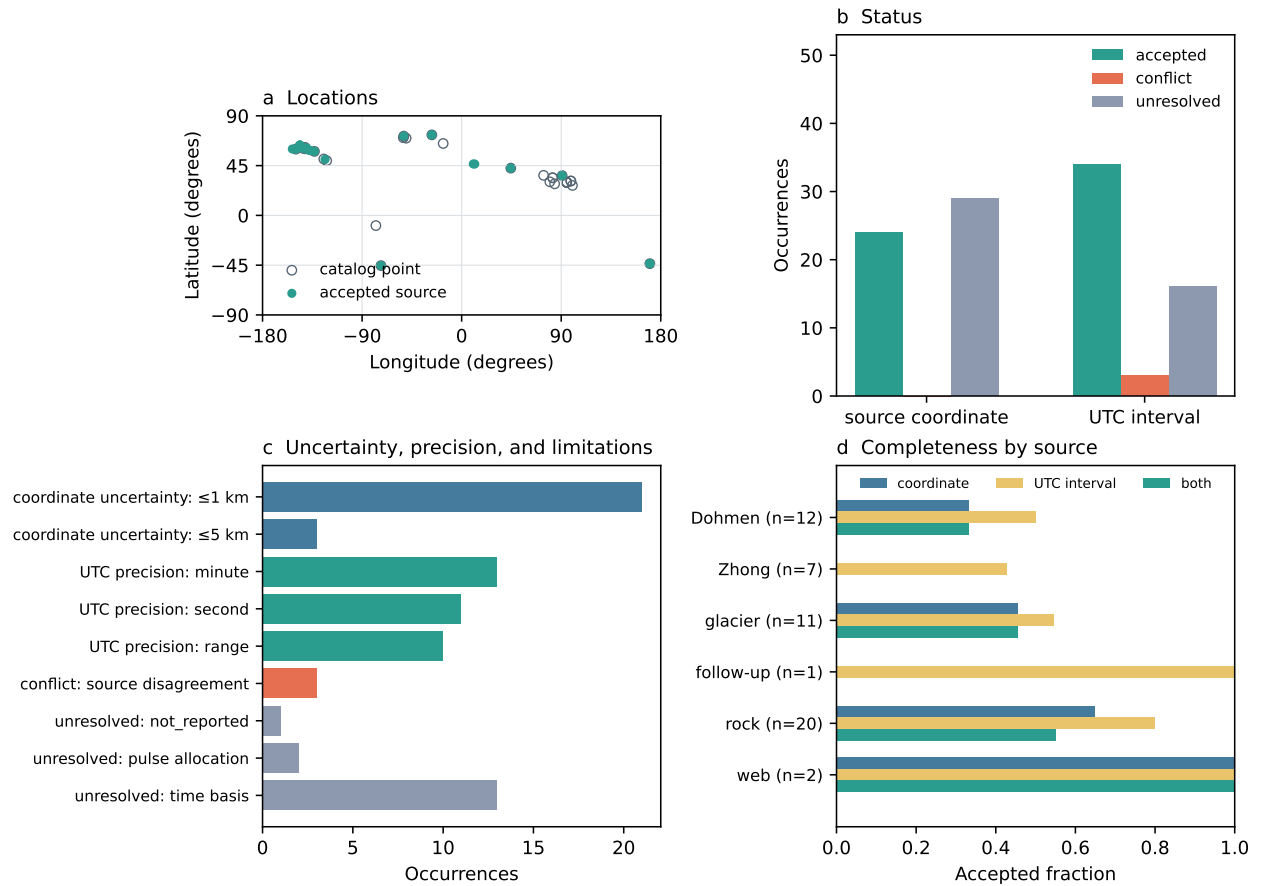


Figure 9: Climate-blind source-coordinate and UTC-onset audit. (a) The 29 finite original catalog points and 24 accepted source coordinates. (b) Accepted, conflicting, and unresolved outcomes. (c) Accepted coordinate uncertainty, accepted UTC precision, and reasons for non-accepted times. (d) Acceptance fractions by discovery stream. Nine accepted UTC intervals are earthquake trigger-origin proxies, not detachment clocks. The figure contains no climate or reanalysis values.

full UTC day earlier. Each mean receives a midrank among matched 1991–2020 calendar windows, excluding the event year. This warm-state rank contains secular change and weather. A separate linear-trend-residual rank removes a Theil–Sen line fitted to matched windows from 1979 through 2025. Nonlinear secular change, low-frequency variability, and reanalysis inhomogeneity can remain. Neither rank estimates failure probability because the sample has no non-failing slopes.

The median warm-state rank is 0.586 (interquartile range 0.379–0.828), with 19 of 29 values above 0.5 (Figure 10). The corresponding median temperature difference from the calendar-control median is 0.486 K (interquartile range -0.508 – 1.414 K). The median local Theil–Sen trend is 0.353 K per decade (interquartile range 0.282–0.509 K per decade). After detrending, the median linear-trend-residual rank is 0.483 (interquartile range 0.267–0.828), and 14 of 29 values exceed 0.5. Selecting the registered representative from each of 25 dependence components gives median warm-state and residual ranks of 0.586 and 0.533, respectively.

In the conservative-antecedence window, the occurrence-level median warm-state and linear-trend-residual ranks are 0.586 (interquartile range 0.483–0.800) and 0.533 (0.300–0.759); 21 and 16 values exceed 0.5. The component-representative medians are 0.621 and 0.552. Buffering changes event-year mean temperature by a median of -0.050 K, but the range is -1.101 to 0.965 K. The other registered warm-state/linear-residual median pairs are 0.552/0.448 for two days, 0.724/0.655 for 30 days, and 0.533/0.448 for the catalog-date day. The central rank is therefore window-dependent. None of these summaries is a hypothesis test or a separation of weather from every slow climate component.

Grid representation is a material measurement uncertainty (Figure 11). The four-cell model-surface height range has a median of 323 m and reaches 1,489 m at Chamoli. Across these cells, the median warm-state-rank range is 0.133 (interquartile range 0.069–0.167) and the maximum is 0.333; the median range of the temperature difference itself is 0.716 K. The buffered four-cell rank range has a median of 0.103 and the same maximum of 0.333. A fixed lapse correction cancels from ranks because it is applied to both event and control values. It does not cancel from absolute 2-m air-temperature hours above 273.15 K. At model elevation, the medians are 113 of 168 hours in the pre-date window and 121 in the buffered window. At the +1 km elevation-offset scenario, lapse rates of 4.0, 6.5, and 9.0 K km $^{-1}$ give pre-date medians of 42, 11, and 0 hours and buffered medians of 50, 11, and 0 hours. These air-temperature calculations do not imply thaw of rock, soil, or ice, and the offsets are scenarios rather than reconstructed source elevations.

6 Competing hypotheses and discriminating observations

The analysis will evaluate signs and timing that distinguish mechanisms. Debuttressing predicts a deformation change point after glacier thinning or contact loss, with a geometry-dependent lag. Loading has no general sign. For an added load Δq on a planar slope, the simple components are $\Delta\tau = \Delta q \sin \alpha$ and $\Delta\sigma_n = \Delta q \cos \alpha$; the Coulomb tendency $\Delta\tau - \Delta\sigma_n \tan \phi$ changes sign with slope and friction angle, before crack pressure and three-dimensional geometry are considered. Hydrologic weakening predicts velocity or failure probability increasing with water input when drainage is inefficient; hydrologic strengthening predicts a negative relation or seasonal hysteresis as drainage evolves. Permafrost degradation predicts deformation associated with cumulative rock-temperature history and ice-bearing fractures rather than a single warm day. Progressive damage predicts accelerating deformation or microseismicity without requiring unusual weather. Structural release predicts failure along mapped adverse discontinuities. Seismic forcing predicts temporal coincidence with an earthquake capable of producing materially large shaking at the source.

Barry Arm provides a prospective case for checking several predictions. Change points and lags

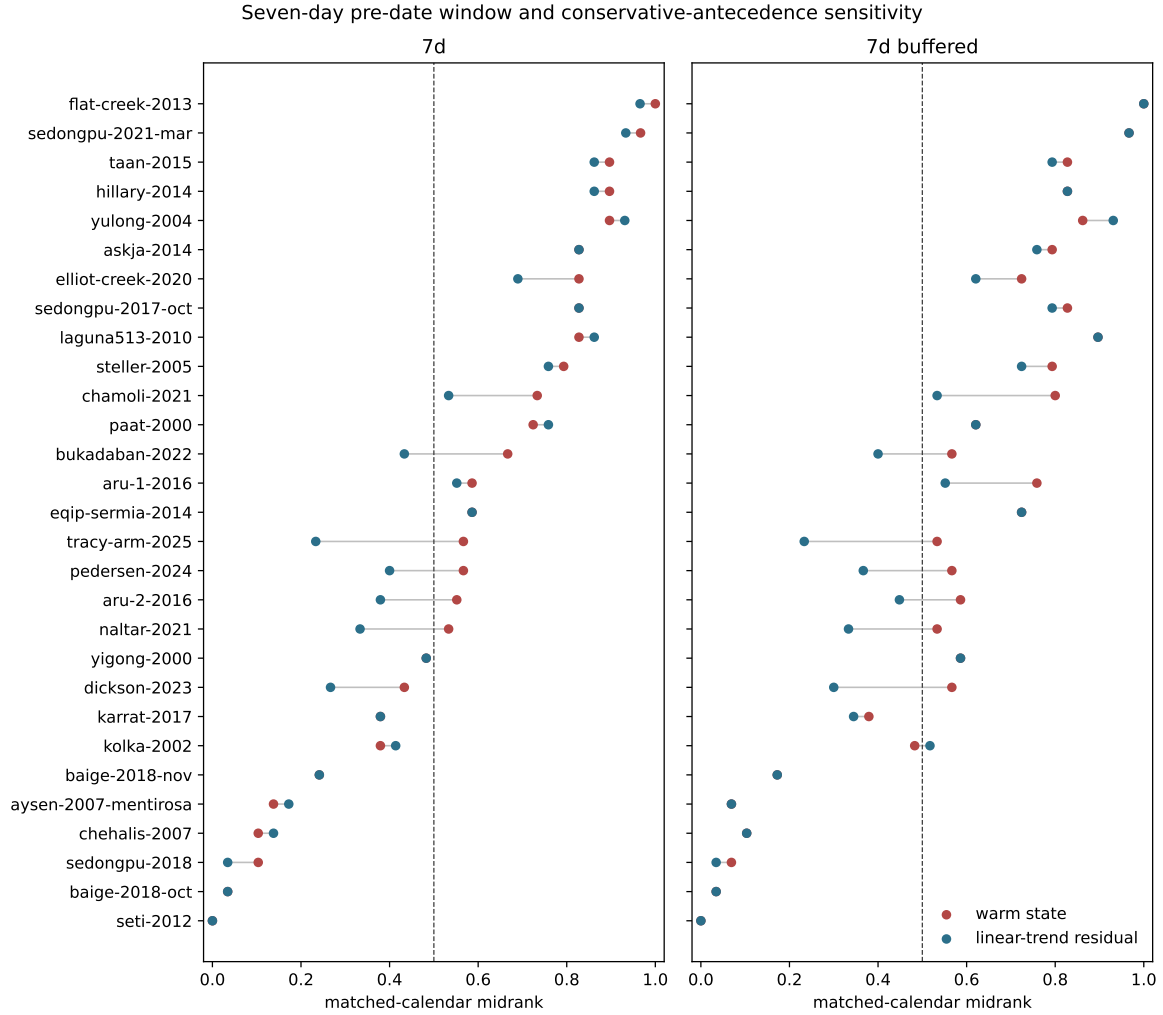


Figure 10: Registered seven-day ERA5 temperature ranks for the pre-date window (left) and conservative-antecedence sensitivity (right), using a common occurrence order. Red points rank event-year means against matching 1991–2020 calendar windows; blue points apply the same rank after removing a robust 1979–2025 linear trend. Lines join two diagnostics for one occurrence. The dashed line at 0.5 is a reference median, not a fitted failure threshold.

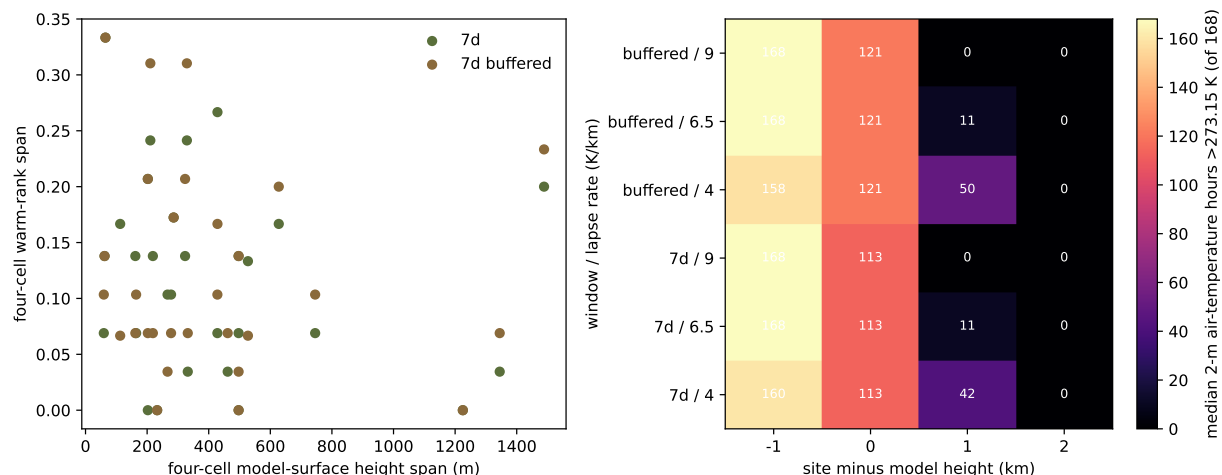


Figure 11: Registered topographic sensitivity. The left panel compares the four-cell model-surface-height range with the corresponding warm-state-rank range for both seven-day windows. The right panel gives the median number of 2-m air-temperature hours above 273.15 K for every window, lapse-rate, and site-minus-model-height scenario. It does not diagnose material thaw, and the scenarios are not estimates of source elevation.

in slope velocity, glacier-contact loss, precipitation, groundwater, and seasonal temperature can be tested with autocorrelated errors. If a glacier term loses explanatory power after detrending or after internal creep state is included, the cumulative 0.99 correlation should not be treated as evidence for a dominant causal pathway.

7 Uncertainty and next steps

Source uncertainty enters at three levels. First, reported intervals are not interchangeable. Chamoli gives a 95% reconstruction interval; Aru reports a conservative linear sum of DEM and post-event correction errors; the probability semantics of some reported Dickson and Tracy uncertainties remain unspecified and are labeled **reported_unspecified**. The Barry Arm depth spread describes agreement among tested parameterizations and time periods, not full inversion nonuniqueness. Second, source classifications can change. The reviewed Nepal seismic record now calls the source a glacial collapse and debris flow, whereas the dated event page retains broader provisional wording [24, 25]. Both records are preserved. Third, some quantities are conditional scenarios. Barry Arm volumes and tsunami heights describe possible geometries, not observations or forecasts of a failure time.

This version completes the climate-blind frame reconciliation, double screening, registered ERA5 temperature pilot, independently reviewed source-coordinate and UTC-onset audit, the first case-blind susceptible- terrain pilot and held-out susceptible-area convergence test, a post-outcome scale-explicit development calculation, and its registered blind numerical transfer test. The audit leaves 29 coordinates and 16 times unresolved and preserves three time conflicts; none is silently removed. The pilot inspected only its registered temperature endpoints and is not retroactively recomputed on the accepted audit subset. A later extraction may use the 22 occurrences with both accepted fields only under a new frozen specification, while retaining all unresolved rows in the reporting frame. The terrain pilot supplies candidate denominator objects, but its three

windows do not estimate regional populations and its connected counts remain scale sensitive. The deterministic held-out windows likewise do not estimate a global population, and both registered area strata fail their resolution criterion. The composite reduces departure in five of six paired development diagnostics. Its unchanged blind calculation then passes all eight registered checks in four hash-selected windows. This closes a numerical transfer test, but the windows have no geographic inclusion probabilities and the strata have not been tested against failure observations. The subsequent geographic design freezes 1,826 RGI 7-intersecting reporting cells and selects 96 with known probabilities. None of the selected cells crosses an RGI-region boundary, a 1.21% chance outcome retained without redraw. The frozen sources replay exactly, but the coverage-only gate stops at a selected p00 cell with no complete DEM-supported reporting center. No equivalent area is calculated; therefore neither complete terrain-screen total is estimable and no partial total is reported. The subsequent full-frame metadata enumeration finds core GLO-90 objects for all 1,826 cells and core GLO-30 objects for 1,822, including the same regional gap, but only 1,422 and 1,411 cells have their complete nine-object halos. Object listing alone neither repairs the stopped design nor establishes valid elevation coverage. A subsequent RGI-geometry screen places 1,065 of 1,073 absent GLO-30 cell-object incidences and 1,030 of 1,031 absent GLO-90 cell-object incidences outside a conservative glacier-proximity screen. The remaining 8 and 1 incidences, including all four absent GLO-30 core cell-object incidences, remain unresolved; geometric exclusion does not establish DEM computability. Physical validation and any later association analysis also require matched non-failing slopes, glacier-contact histories, deformation records, permafrost proxies, additional meteorological variables, and a frozen statistical specification. Only then can the study estimate whether hazardous alpine mass movements occupy systematically different climatic or cryospheric conditions.

8 Conclusions

The catalog establishes a traceable distinction among initial failure, downstream cascade, climate preconditioning, and immediate trigger. The climate-blind screen retains 73 eligible occurrences while keeping 68 unresolved cases visible. It also shows why reanalysis alone is insufficient: glacier detachments require effective-pressure bounds, while rock-slope cases require glacier geometry, support change, structure, and deformation. In the registered 29-occurrence pilot, central ranks change with the window and fitted-linear-trend adjustment. Neighboring cells change ranks and anomalies, while elevation scenarios change above-freezing air-temperature counts. The subsequent audit accepts source coordinates for 24 of 53 eligible records, UTC intervals for 34, and both for 22. Only 9 of the pilot's 29 occurrences have accepted source coordinates, so its exposure summaries describe registered catalog locations rather than a source-verified sample. Coordinate and time acceptance are themselves source-dependent, and their representativeness is unknown. The blind terrain pilot reproducibly enumerates source-like area, but 30 versus 90 m grids change primary connected counts by factors of 1.3–3.0. In the held-out area test, phase coefficients of variation remain below 0.004, but 90 m departures reach 26.7–36.6% for glacier-contact area and 29.1–37.4% for PZI-intersection area in Arctic Canada and Svalbard; both strata fail. A post-outcome composite gives 2.6–13.4% departures in the same windows and is lower in five of six paired diagnostics. Its unchanged blind calculation gives 1.8–5.3% departures and four-phase coefficients of variation no greater than 0.00375 in four additional hash-selected windows; all eight registered checks pass. This establishes numerical transfer only. The windows are not a probability sample, and the calculation has not been tested against failing and non-failing slopes. A later signed-beacon draw supplies a separate 96-cell probability sample from 1,826 RGI 7-intersecting cells. Its frozen-source coverage gate stops before equivalent-area execution because one selected cell has zero complete

DEM support. The complete design-based glacier-outline-proximity and PZI-intersection totals are therefore non-estimable from that source; a zero or partial estimate would violate the design. A full-frame object audit subsequently lists all core GLO-90 objects and 1,822 of 1,826 core GLO-30 objects, but only 77.88% and 77.27% of cells have their complete nine-object halos. Because listing does not establish readable and valid elevation values, the subsequent conservative geometry screen is deliberately narrower: it places 1,065 GLO-30 and 1,030 GLO-90 absent cell-object incidences outside the glacier-proximity screen, while 8 and 1 incidences remain unresolved. It neither repairs the registered stop nor proves terrain coverage. A newly frozen, source-valid design is required before those totals, detrended seasonal controls, or explicit competing predictions can be tested. The present contribution is a sampling frame, protocol, and exposure-sensitivity result, not evidence that warming systematically triggered these events.

References

- [1] Airbus Defence and Space. *Copernicus Digital Elevation Model: Product Handbook*, issue 5.0 edition, 2024. Document GEO1988-CopernicusDEM-SPE-002.
- [2] Amazon Web Services. Registry of open data on aws: Copernicus dem. <https://registry.opendata.aws/copernicus-dem/>, 2026. Accessed 31 August 2026.
- [3] Katherine R. Barnhart, Ryan P. Jones, David L. George, Jeffrey A. Coe, and Dennis M. Staley. Preliminary assessment of the wave generating potential from landslides at Barry Arm, Prince William Sound, Alaska. Open-File Report 2021-1071, U.S. Geological Survey, 2021.
- [4] Andrew L. Collins, Kate E. Allstadt, and Dennis M. Staley. Using the horizontal-to-vertical spectral ratio method to estimate thickness of the Barry Arm landslide, Prince William Sound, Alaska. Open-File Report 2024-1071, U.S. Geological Survey, 2024.
- [5] Chunli Dai, Bretwood Higman, Patrick J. Lynett, Mylène Jacquemart, Ian M. Howat, Anna K. Liljedahl, Anja Dufresne, Jeffrey T. Freymueller, Marten Geertsema, Melissa Ward Jones, and Peter J. Haeussler. Detection and assessment of a large and potentially tsunamigenic periglacial landslide in Barry Arm, Alaska. *Geophysical Research Letters*, 47(22), 2020. doi: 10.1029/2020GL089800.
- [6] Katrin Dohmen, Anika Braun, and Tomas Manuel Fernandez-Steege. A catalog of landslide-triggered tsunamis, 2025.
- [7] M. D. Dokukin. Ice and rock avalanches in the central caucasus in 2025. *Ice and Snow*, 66(3): 476–485, 2026. doi: 10.7868/S2412376526030053.
- [8] Earthmover. Icechunk ERA5: analysis-ready cloud-optimized reanalysis. AWS Registry of Open Data, 2026. URL <https://registry.opendata.aws/earthmover-era5/>. Snapshot T9H8SG2PVXWNY0QNJJPJG; accessed 30 August 2026.
- [9] European Centre for Medium-Range Weather Forecasts. ERA5 reanalysis (0.25 degree latitude–longitude grid), 2019. Updated monthly.
- [10] European Space Agency and European Union. Copernicus digital elevation model, glo-30, 2021 release. Copernicus Data Space Ecosystem and AWS Registry of Open Data, 2021.

- [11] Alex S. Gardner, Mark Fahnestock, Chad A. Greene, Jonathan H. Kennedy, M. Liukis, L. López, and Ted A. Scambos. MEaSUREs ITS_LIVE regional glacier and ice sheet surface velocities, version 2, 2025.
- [12] Adrien Gilbert, Silvan Leinss, Jeffrey Kargel, Andreas Kääb, Simon Gascoin, Gregory Leonard, Etienne Berthier, Alina Karki, and Tandong Yao. Mechanisms leading to the 2016 giant twin glacier collapses, Aru Range, Tibet. *The Cryosphere*, 12(9):2883–2900, 2018. doi: 10.5194/tc-12-2883-2018.
- [13] Stephan Gruber. Derivation and analysis of a high-resolution estimate of global permafrost zonation. *The Cryosphere*, 6:221–233, 2012. doi: 10.5194/tc-6-221-2012.
- [14] Stephan Gruber. Global permafrost. Edinburgh DataShare dataset, CC BY 3.0, 2017.
- [15] Hans Hersbach et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi: 10.1002/qj.3803.
- [16] Andreas Kääb et al. Massive collapse of two glaciers in western Tibet in 2016 after surge-like instability. *Nature Geoscience*, 11(2):114–120, 2018. doi: 10.1038/s41561-017-0039-7.
- [17] Peter Mani, Simon Allen, Stephen G. Evans, Jeffrey S. Kargel, Martin Mergili, Dmitry Petrakov, and Markus Stoffel. Geomorphic process chains in high-mountain regions—a review and classification approach for natural hazards assessment. *Reviews of Geophysics*, 61(4):e2022RG000791, 2023. doi: 10.1029/2022RG000791.
- [18] Thomas Monahan, Tianning Tang, Stephen Roberts, and Thomas A. A. Adcock. Observations of the seiche that shook the world. *Nature Communications*, 16:4777, 2025. doi: 10.1038/s41467-025-59851-7.
- [19] RGI 7.0 Consortium. Randolph glacier inventory – a dataset of global glacier outlines, version 7.0, 2023.
- [20] L. N. Schaefer et al. Kinematic evolution of a large paraglacial landslide in the Barry Arm Fjord of Alaska. *Journal of Geophysical Research: Earth Surface*, 128(11), 2023. doi: 10.1029/2023JF007119.
- [21] D. H. Shugar et al. A massive rock and ice avalanche caused the 2021 disaster at Chamoli, Indian Himalaya. *Science*, 373(6552):300–306, 2021. doi: 10.1126/science.abh4455.
- [22] Dan H. Shugar et al. A 481-meter-high landslide-tsunami in a cruise ship-frequented Alaska fjord. *Science*, 392(6803), 2026. doi: 10.1126/science.aec3187.
- [23] Kristian Svennevig et al. A rockslide-generated tsunami in a Greenland fjord rang Earth for 9 days. *Science*, 385(6714):1196–1205, 2024. doi: 10.1126/science.adm9247.
- [24] U.S. Geological Survey Earthquake Hazards Program. M 5.2 landslide—55 km northwest of Kodari, Nepal. Comprehensive Catalog event us7000tbwb, 2026. URL <https://earthquake.usgs.gov/earthquakes/eventpage/us7000tbwb/executive>. Accessed 30 August 2026.
- [25] U.S. Geological Survey Landslide Hazards Program. 2026 Nepal debris avalanche and flash flood. Event summary, findings as of 27 August 2026, 2026. URL <https://www.usgs.gov/programs/landslide-hazards/science/2026-nepal-debris-avalanche-and-flash-flood>. Accessed 30 August 2026; preliminary and subject to revision.

- [26] Maximillian Van Wyk de Vries, Shashank Bhushan, Mylène Jacquemart, César Deschamps-Berger, Etienne Berthier, Simon Gascoin, David E. Shean, Dan H. Shugar, and Andreas Kääh. Pre-collapse motion of the February 2021 Chamoli rock–ice avalanche, Indian Himalaya. *Natural Hazards and Earth System Sciences*, 22(10):3309–3327, 2022. doi: 10.5194/nhess-22-3309-2022.
- [27] Yan Zhong, Simon Allen, and Markus Stoffel. Supplementary data to: Large rock and ice avalanches frequently produce cascading processes in High Mountain Asia, 2024.
- [28] Yan Zhong, Simon Keith Allen, Guoxiong Zheng, Qiao Liu, and Markus Stoffel. Large rock and ice avalanches frequently produce cascading processes in High Mountain Asia. *Geomorphology*, 449:109048, 2024. doi: 10.1016/j.geomorph.2023.109048.